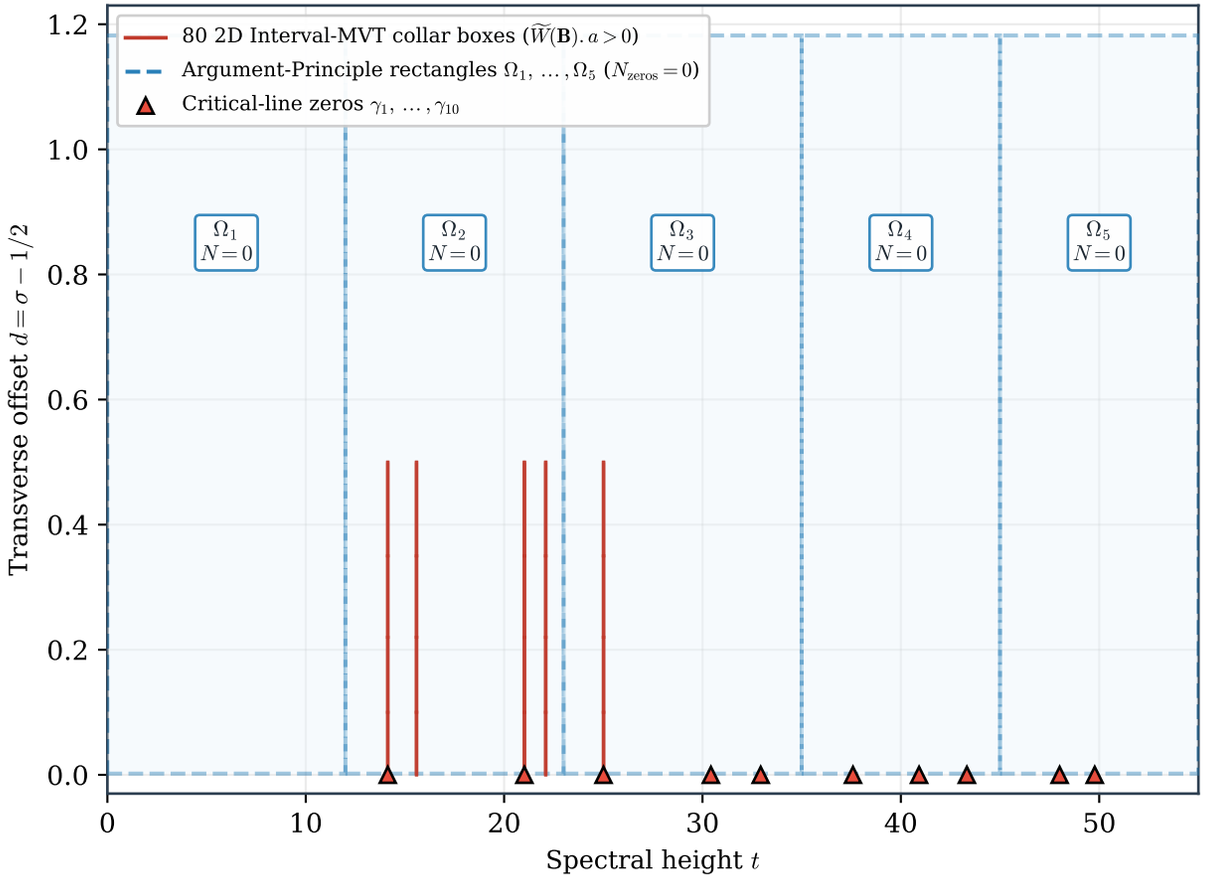
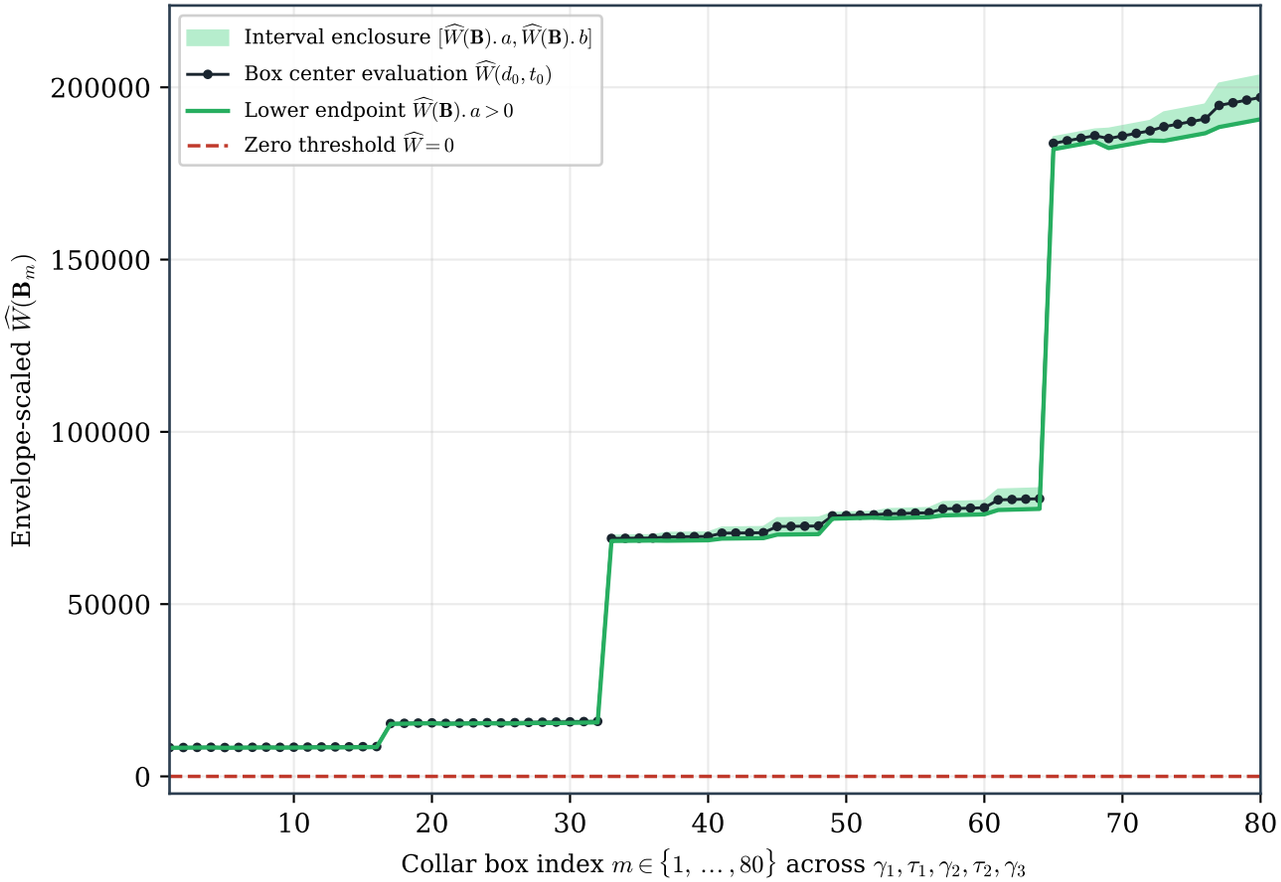
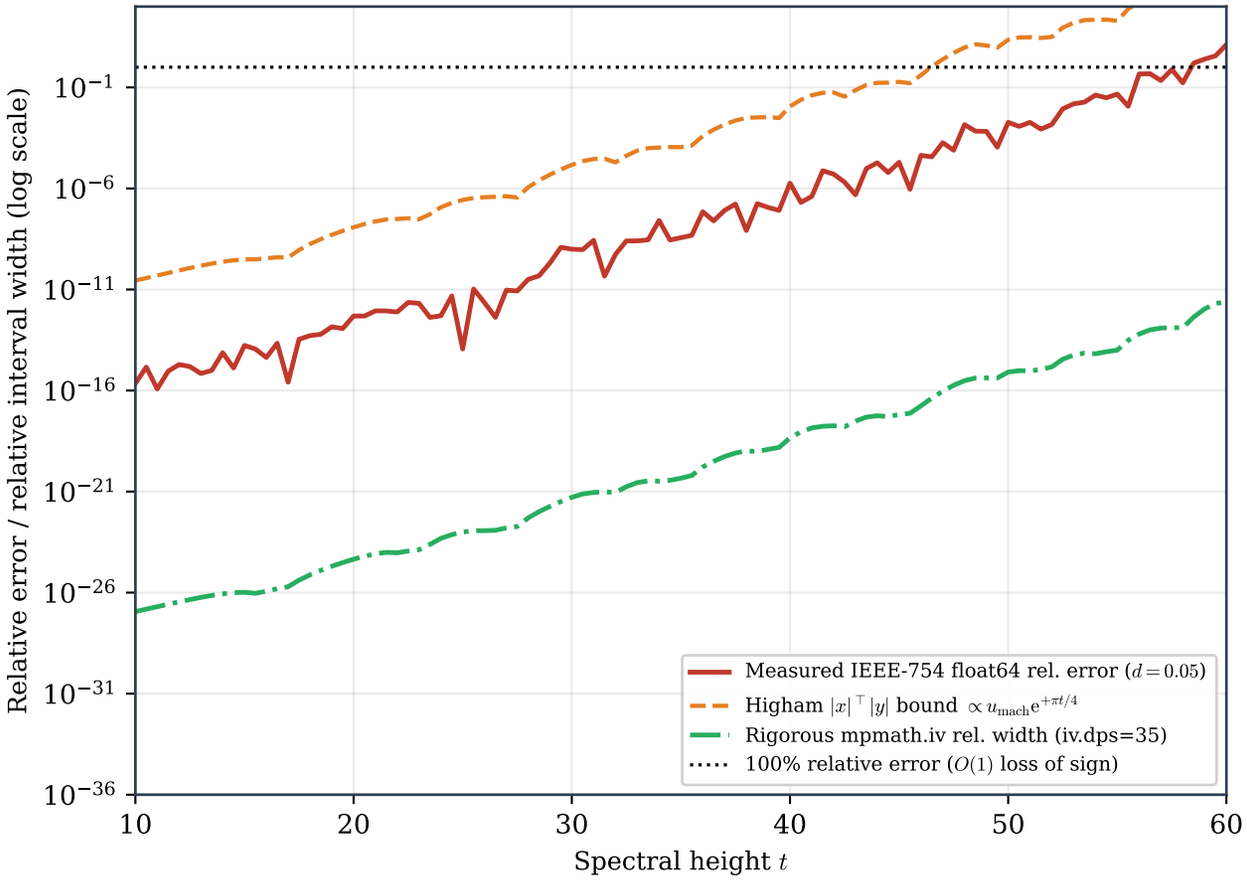
(a) Benchmark Partition: Collar Boxes (γ_k, τ_k) and Rectangles $\Omega_1, \ldots, \Omega_5$



(b) Non-Harmonic-Safe 2D Interval-MVT Enclosures (80 Collar Boxes)



(c) IEEE-754 float64 Cancellation Breakdown vs. mpmath.iv



(d) Integer Argument Principle Enclosures $N_{\text{zeros}}(\Omega_k) \in (-0.5, 0.5) \cap \mathbb{Z} = \{0\}$

